Local versus Global: Comparison of Convolutional and Transformer Models for Skin Lesion Classification with an Emphasis on Resources

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***Abstract*—**

***Index Terms*—computational efficiency, convolutional neural networks, dermoscopy, HAM10000, skin lesion classification, vision transformers**

I. Introduction

Deep neural networks can easily match dermatologists on skin cancer classification tasks [1], which has made automated image analysis a route to efficient and cost-effective detection. Publicly available datasets such as HAM10000 [2] make it possible to train and compare such models, but class imbalance, limited data, and image selection bias [3] make reliable evaluation difficult in real-life conditions.

Accuracy alone also gives an incomplete comparison due to class imbalance and the unequal severity of misclassifications. Lesion segmentation masks do not consistently improve classification, and removing the background context can even reduce accuracy [6], so the value of mask information depends on how it is used.

In this project, we will compare local and global models for seven-class skin lesion classification on HAM10000, with an emphasis on the resources each model requires. ResNet-18, ResNet-34, and ResNet-50 represent convolutional models of increasing size, and ViT-S/16 represents the transformer family. We will also evaluate a U-Net-based approach with a ResNet-34 encoder and a classification head, and investigate mask-assisted classification.

II. Relevant Literature

The potential of deep learning for skin cancer classification was demonstrated by Esteva et al. [1], who trained a convolutional neural network (CNN) on 129,450 clinical images covering 2,032 skin diseases. Their model achieved performance comparable to dermatologists on selected skin cancer classification tasks, highlighting the potential for accessible and cost-effective skin cancer detection.

The HAM10000 dataset, introduced by Tschandl et al. [2], contains 10,015 dermoscopic images across seven diagnostic categories. It provides a foundation for training and evaluating skin lesion classification models. However, its class imbalance presents a challenge when evaluating model performance.

A systematic review conducted by Kassem et al. [3] examined 53 machine learning and 49 deep learning methods for skin lesion classification. The authors identified major challenges such as limited datasets, image selection bias, and variations in skin color. These findings underscore the importance of reliable evaluation methods and appropriate data preprocessing.

Several studies have investigated the performance of CNNs and vision transformers (ViTs) for skin lesion classification. One such study [4] compared ResNet-152 and ViT on the HAM10000 dataset. ViT achieved an accuracy of 98.28%, compared to 96.70% for ResNet-152. Although ViT achieved higher classification accuracy, the authors also noted its computational requirements and risk of overfitting.

A broader comparison of CNN and transformer-based models was conducted by Aruk et al. [5] using the HAM10000 and ISIC 2019 datasets. Their findings showed that transformer-based architectures can achieve competitive classification performance and that larger models do not necessarily outperform smaller ones. The study also investigated the effects of data augmentation and transfer learning.

The influence of lesion segmentation on classification performance was investigated by Mahbod et al. [6]. Their results showed that segmentation masks do not consistently improve classification performance. Removing the background could even reduce accuracy by discarding useful contextual information.

In addition to classification accuracy, computational efficiency is another important factor when comparing deep learning models. Ecer et al. [7] compared ResNet-50, EfficientNet-B3, and ViT-B/16 on the HAM10000 dataset. All three models achieved similar accuracy, ranging from 80.7% to 81.7%. However, ViT required substantially more energy, suggesting that higher computational costs do not necessarily lead to better classification performance.

Finally, a framework combining segmentation and classification was proposed by Fiaz et al. [8]. The authors used U-Net with a ResNet-34 encoder for skin lesion segmentation and EfficientNet-B0 for classification. On the HAM10000 dataset, their model achieved 85.57% classification accuracy

Fig. 1. Random sample images per class, one image per lesion.

Fig. 2. Label distribution and the share of each class contributed by each source.

and a Dice score of 0.8622 for segmentation. These results demonstrate the potential of combining segmentation and classification for skin lesion analysis.

III. Data Analysis

We use HAM10000 [2], obtained from the Harvard Dataverse. It was collected to train and benchmark automated diagnosis of pigmented skin lesions and contains 10,015 images of 7,470 lesions from four sources in Austria and Australia. Inputs are 600×450 RGB dermoscopic images (Fig. 1) with binary lesion masks, age, sex, and body site. Labels are seven diagnoses (Table I), confirmed by histopathology for 53.3% of images and otherwise by follow-up, expert consensus, or confocal microscopy. Classes range from 66.9% of images for nv to 1.1% for df, a 58:1 imbalance (Fig. 2).

The data will be divided into training and validation folds grouped by lesion, because 44.9% of images show lesions photographed more than once.

The RGB channel means [0.764, 0.546, 0.571] differ substantially from those of ImageNet, so ImageNet statistics will be used only for pretrained models. Image properties differ strongly between sources (Table II).

Source is correlated with the diagnosis: `vidir_molemax` is 94.1% nv and 0.6% mel, and differs visibly in appearance from the other sources (Fig. 2). Class-level trends can reverse within a source (Table I): saturation is higher for nv than for mel overall, but lower in three of four sources, so a model may learn the source rather than the lesion itself.

Heuristically detected dark corners (vignettes) affect 26.2% of mel but only 7.8% of nv images, and 32.2% of `vidir_modern` images; hair affects 12.6% of images. The most extreme images in sharpness and exposure were inspected visually.

We expect vasc to be easily detectable (distinctive red hue, $\Delta E \approx 32.3$), while low-contrast akiec and bcc ($\Delta E \approx 11$), mel versus nv, and rare df should be harder. Training will use crops, flips, rotations, mild color jitter guided by source differences, and a class-independent vignette in order to avoid data poisoning by existing ones.

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TABLE I

IMAGES, LESIONS, MEDIAN AGE PER LESION, AND MEDIAN SATURATION AND LESION-TO-SKIN ΔE PER IMAGE, BY CLASS AND SOURCE (ROS: ROSENDAHL, MOD: VIDIR_MODERN, MOL: VIDIR_MOLEMAX, DIAS: VIENNA_DIAS)

				Saturation					ΔE				
Class	Images	Lesions	Age	All	Ros	Mod	Mol	Dias	All	Ros	Mod	Mol	Dias
Melanocytic nevus (nv)	6705	5403	45	.360	.210	.198	.423	.313	26.6	27.0	33.0	24.6	31.8
Melanoma (mel)	1113	614	65	.250	.262	.235	.387	.337	28.1	23.1	29.7	32.3	31.9
Benign keratosis (bkl)	1099	727	65	.226	.209	.219	.372	.281	16.2	13.2	19.0	18.3	23.9
Basal cell carcinoma (bcc)	514	327	70	.215	.223	.206	–	–	11.0	9.8	12.9	–	–
Actinic keratosis / IEC (akiec)	327	228	67.5	.264	.265	.250	–	–	10.9	10.9	10.5	–	–
Vascular lesion (vasc)	142	98	55	.249	–	.212	.330	–	32.3	–	30.4	33.0	–
Dermatofibroma (df)	115	73	55	.245	.218	.218	.351	–	15.0	11.0	18.5	14.0	–
All	10015	7470	50	.291	.227	.211	.420	.320	24.8	17.3	28.8	24.5	31.5

TABLE II

PER-IMAGE PROPERTIES

Property	Mean	Med	Min	Max	Std	Ros	Mod	Mol	Dias
Brightness	.765	.774	.323	.997	.097	.751	.693	.839	.753
Contrast	.109	.101	.019	.335	.045	.088	.115	.096	.144
Saturation	.306	.291	.035	.710	.118	.227	.211	.420	.320
Sharpness	1.89	1.83	0.94	3.68	0.34	1.93	1.82	1.76	2.10
Near-black	3.4	–	–	–	–	1.8	8.6	0.0	3.2